Introduction to Responsible AI

03

CSG202 · Module 3

Learning objectives

Explain responsible AI

Describe why accuracy alone is not enough for a trustworthy AI system.

Map stakeholder impact

Identify who benefits, who may be harmed, and who can challenge a decision.

Trace lifecycle decisions

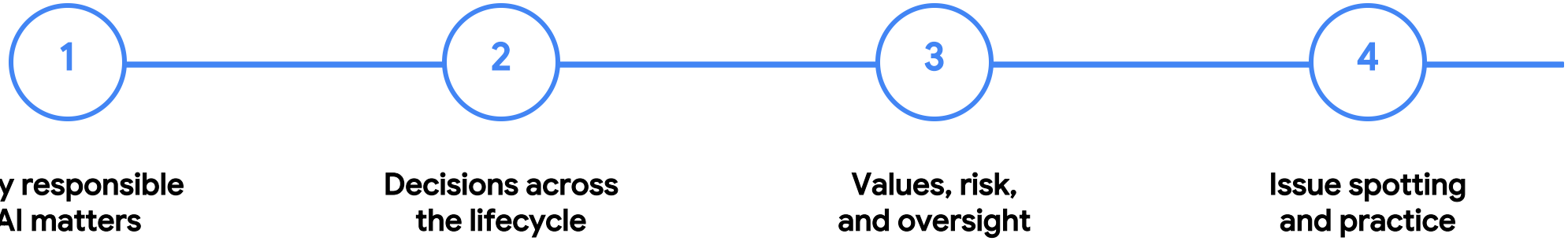
Recognize responsible choices from problem framing through monitoring.

Connect values to controls

Translate a value such as fairness into a requirement, test, and evidence.

Module topics

A connected journey from intent to accountable operation



01



Why responsible AI matters

AI systems interact with people, policies, data, and real-world consequences

An AI system is part of a socio-technical system

The model is only one component of the outcome



Look beyond the model

- People decide goals, labels, thresholds, and responses.
- Policies define acceptable use and escalation.
- Data reflects past choices and missing voices.
- Interfaces shape what users understand and do next.

Responsible AI connects capability with consequences

Can the system do the task?

Technical evaluation

- Quality and reliability
- Latency and cost
- Robustness under variation

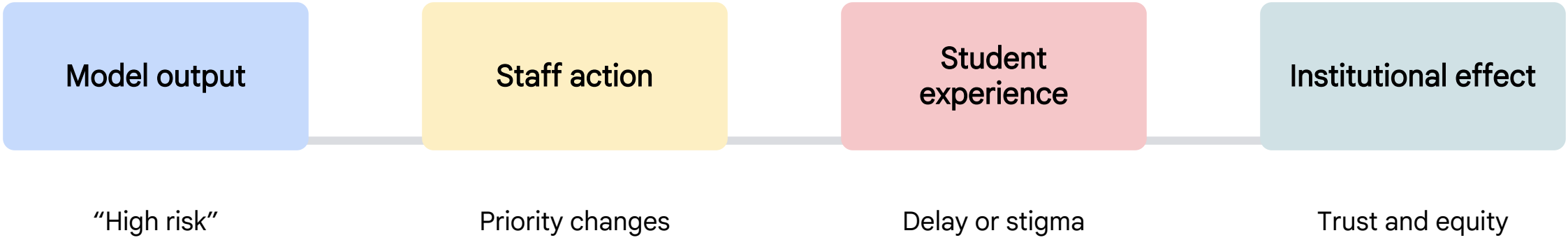
Should it be used this way?

Responsible evaluation

- People and rights
- Context and power
- Accountability and recourse

A trustworthy solution needs both answers.

One model output can create a chain of effects



Responsible AI examines the entire chain—not only the first box.

Scenario: an AI assistant routes student support requests

The proposal

Summarize incoming messages and assign urgency so counselors can respond faster.

Potential benefit

Shorter queues and faster attention for urgent needs.

Questions before launch

- Are mental-health messages allowed in training or logging?
- Could language, disability, or culture affect urgency scores?
- Can a student reach a person without using the assistant?
- Who reviews low-confidence or high-risk cases?
- How will errors be reported and corrected?

Use case + context + safeguards = a responsible design conversation

Responsible design balances benefit, likelihood, and severity

Expected benefit

Who benefits, how much, and how will we know?

Potential harm

Who may be affected, how likely, and how severe?



Risk is not “zero or unacceptable.” Controls should be proportional to context.

Five recurring questions organize a responsible AI review

People
Who is affected?

Data
What is represented?

Is it useful
for the stated need?

Behavior
When can it fail?

Evidence

How will claims be tested?

Control
Who can intervene?

Use case: AI-generated feedback on student writing

Goal

Give faster formative feedback while preserving teacher judgment.

Useful output

Highlights unclear claims and suggests questions—not a final grade.

Risk scenario

Dialect or non-standard English is treated as lower-quality reasoning.

Controls

Teacher review, rubric-based evaluation, student appeal, and subgroup checks.

Design choice: support learning, do not silently replace the accountable educator.

The same AI capability can require different safeguards

Dimension	Retail support	Healthcare
Context	Customer-service chat	Clinical consultation
Possible harm	Incorrect refund or delay	Missed symptom or unsafe advice
Oversight	Agent reviews before sending	Qualified clinician verifies
Recourse	Escalation and complaint	Immediate clinical escalation

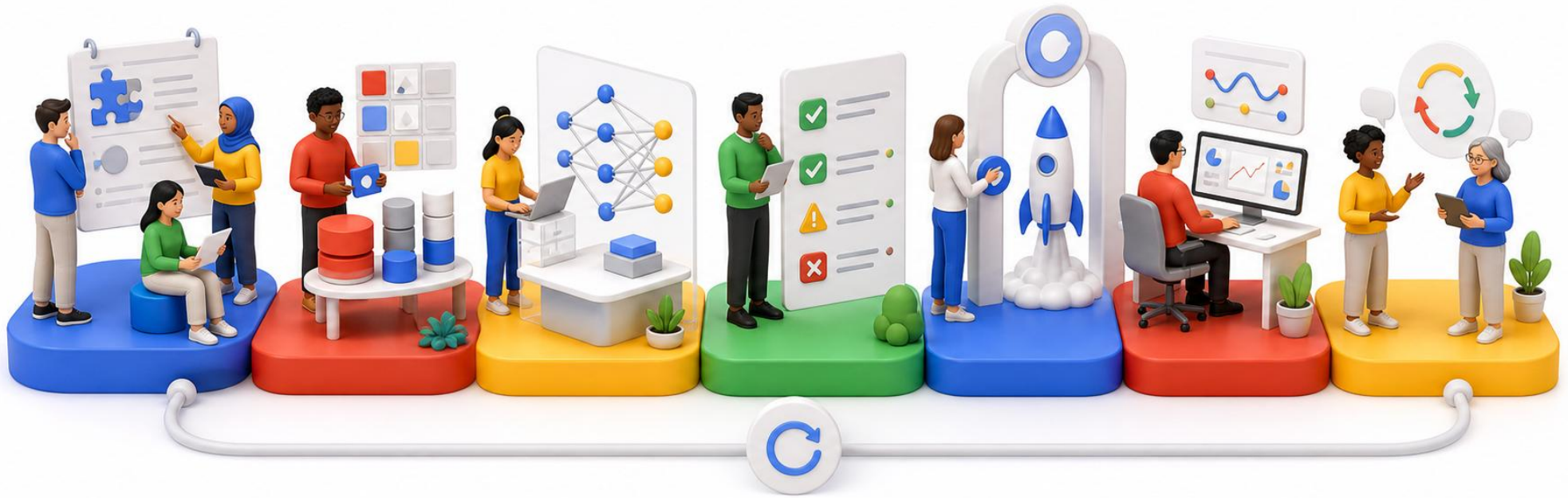
Risk depends on the decision context—not only the model.

02

Responsible decisions across the lifecycle

Choices before training, during evaluation, at launch, and after deployment

Responsible AI is a continuous lifecycle



The feedback loop updates the problem, data, tests, and controls.

Each stage asks a different question

Frame	Data	Build	Evaluate	Launch	Monitor
Should we build it?	Whose reality is represented?	What behavior is optimized?	Does it work safely for this context?	Who can rely on it—and how?	What changed after release?

A missed decision early in the lifecycle is expensive to repair later.

Problem framing: begin with the need, not the model

Weak framing

“Use generative AI to automate student support.”

Clear framing

“Reduce first-response time while preserving counselor control and urgent escalation.”

Scenario test Would a simpler queue rule or form redesign solve the need with less risk?

Data decisions determine what the system can learn

Represent

Which students, languages, and situations appear?

Permission

Was the data collected and reused appropriately?

Quality

Are labels consistent, current, and meaningful?

Protection

What sensitive information must be minimized?

Example: If urgent messages in regional languages are rare in training data, the assistant may under-prioritize them even when overall accuracy looks high.

Model choices should match the consequence of an error

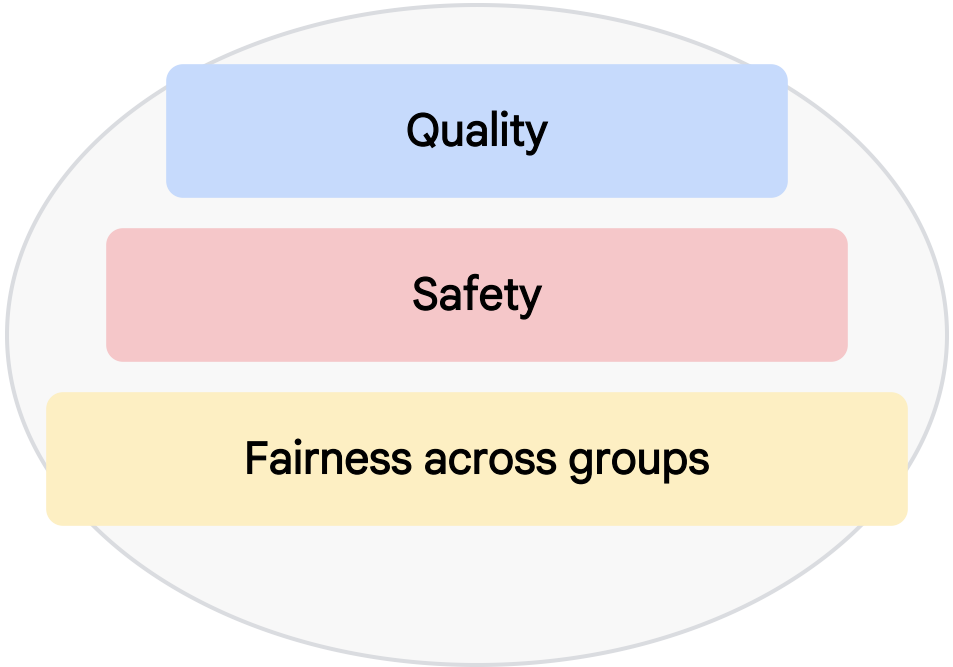
Task	Model and prompt	Decision rule	Human action
What is predicted?	**What evidence is used?**	**What threshold triggers action?**	**Who owns the outcome?**
Urgency category	Message + allowed context	Low confidence → review	Counselor

A model is not the decision: thresholds, interface design, and human response complete the system.

Evaluation needs more than one score

Also test

- Real user tasks
- Edge cases
- Misuse attempts
- Uncertainty



Evidence

- Metrics
- Human review
- User research
- Red-team findings

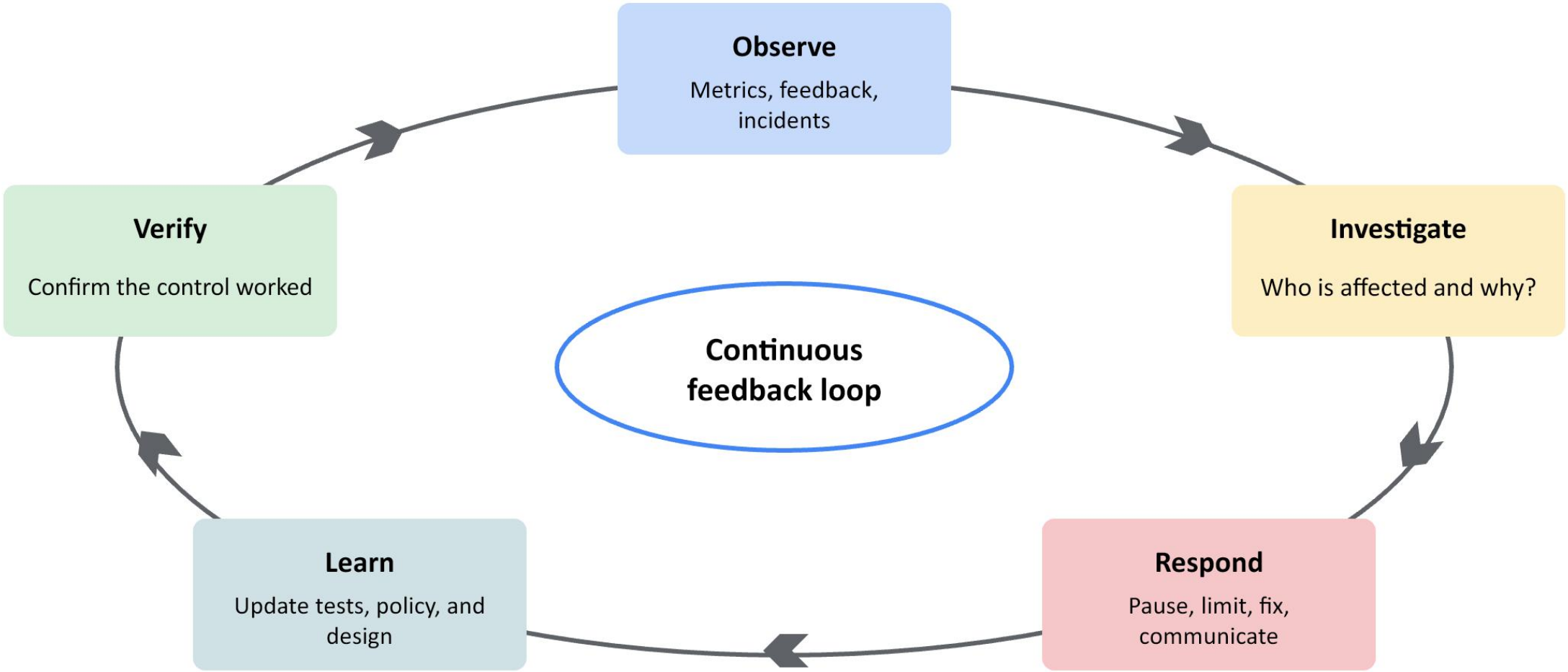
No single metric proves that a system is responsible.

A system can pass testing and still fail after launch

During testing	After launch
Test data Known semester patterns	Live data Exam-season surge
Users Pilot counselors	Users New staff and languages
Behavior Expected requests	Behavior Workarounds and misuse

Deployment changes the environment; monitoring checks whether assumptions remain true.

Monitoring turns signals into action



Scenario: the assistant misses urgent messages in one language

1 Detect

Counselor reports repeated misses

2 Contain

Route all affected messages to manual review

3 Understand

Check data, prompt, threshold, and UI

4 Correct

Improve controls and re-evaluate

5 Communicate

Inform users and document learning

The responsible response protects people first, then diagnoses and improves the system.

03

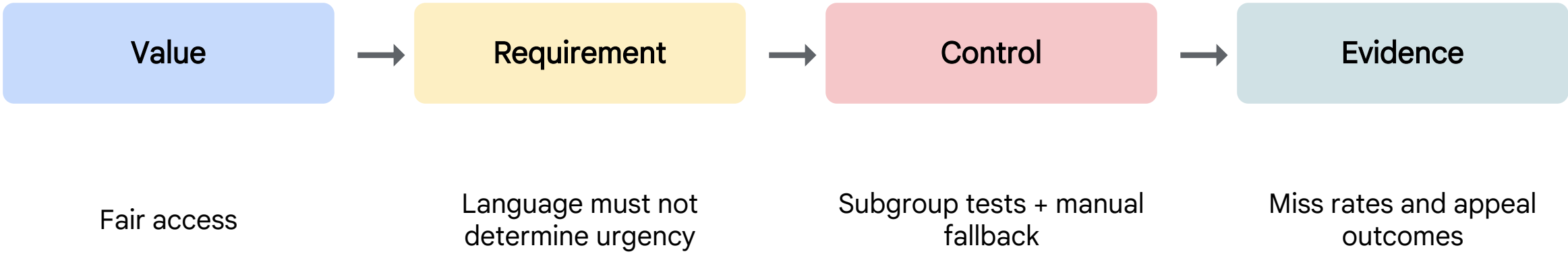
Values, risk, and human oversight

Translate intentions into requirements, controls, evidence, and recourse

Business need and human outcome must be measured together

1	**Need**	Reduce average first-response time
2	**Human outcome**	Urgent students receive timely, appropriate help
3	**Quality measure**	Routing agreement with trained counselors
4	**Safety measure**	Critical-miss rate and escalation time
5	**Experience measure**	Can students understand and challenge the route?

A value becomes useful when it changes a design decision



If a team cannot name the requirement, control, and evidence, the value is still only an aspiration.

Oversight should increase with consequence and uncertainty

Lower	Moderate	Higher
Brainstorm event names	**Draft student FAQ**	**Recommend urgent intervention**
User reviews output	Owner checks facts and tone	Trained reviewer + fallback + appeal

Risk tiering is contextual: the same model may be low-risk in one use and high-risk in another.

Human oversight needs authority, information, and recourse



A human is not a safeguard if they cannot understand, question, or overturn the output.

“Human in the loop” can mean three different arrangements

Human-in-the-loop

A person approves each consequential output before action. Example: counselor confirms urgent routing.

Human-on-the-loop

Automation proceeds, while a person supervises alerts and can intervene. Example: agent watches queue anomalies.

Human-out-of-the-loop

No routine review before action. Suitable only when consequences are low and recovery is easy.

Choose the arrangement from the risk—not from convenience alone.

Transparency should help someone make a better decision

User	Know AI is involved and what to verify
Reviewer	See evidence, uncertainty, and limits
Owner	Understand tests, incidents, and change history

More information is not always more useful—match the explanation to the audience.

Recourse makes a consequential system contestable

Notice	Question	Review	Correct	Learn
Tell the person what happened	Provide a clear route to challenge	A qualified person re-examines	Fix the outcome and relevant data	Use patterns to improve the system

Example: a student can appeal an incorrect urgency category without having to understand the model.

Scenario: should AI recommend scholarship candidates?

Potential value

Help reviewers find evidence across many applications.

High-consequence concerns

- Historical award patterns may encode exclusion.
- Generated summaries may omit context.
- Applicants may not know how to challenge errors.

Responsible redesign

Role	Summarize evidence; never rank or decide
Review	Two trained reviewers inspect source text
Evaluation	Check omission and subgroup error patterns
Recourse	Applicants can correct factual errors

Sometimes responsibility means narrowing the task.

Use case pattern: assist the expert, preserve the accountable decision

User request

AI drafts or organizes

Expert verifies

User receives answer + route to help

Examples

- Student support
- Employee HR FAQ
- Customer service
- Clinical documentation

Key condition

The expert must have source context, time, training, and authority to correct the draft.

04



Issue spotting and practice

Use simple questions, pre-mortems, and worked scenarios before deployment

Use five questions before approving an AI use case

1	**Purpose**	What need is being solved—and is AI necessary?
2	**People**	Who benefits, who may be harmed, and who is missing?
3	**Failure**	What can go wrong, and how serious is it?
4	**Control**	Who can intervene, appeal, pause, or stop?
5	**Evidence**	What tests and monitoring justify trust?

Pre-mortem: imagine the deployment failed six months later

“The system caused harm, lost trust, or was stopped. What happened?”

Failure story	Early signal	Preventive control	Owner
Describe a specific person, event, and consequence.	What would have warned the team sooner?	What design or process could reduce the risk?	Who acts when the signal appears?

A pre-mortem turns vague concern into testable signals and owned controls.

Worked example: responsible campus support assistant

Frame

Faster first response without replacing counselor judgment

Data

Consent, language coverage, sensitive-data minimization

Build

Summarize + suggest urgency; expose confidence

Evaluate

Critical misses, subgroup results, counselor review

Launch

Manual fallback, notice, escalation, limited pilot

Monitor

Drift, appeals, incidents, and periodic re-evaluation

The system remains accountable because each stage has a decision, control, and evidence.

Google Skills activity

Introduction to Responsible AI

What learners do

- Review why responsible AI matters
- Identify common dimensions and practices
- Connect principles to project decisions
- Use the course environment—no personal GCP account required

Trainer setup

- Confirm the activity is currently available
- Explain temporary access or course credits
- Keep the official instructions visible
- Debrief with the five issue-spotting questions

Skills Google course template: 554

Class activity: improve an AI use case

Think · Pair · Share

Choose one proposal:

AI attendance monitoring · automated grading · campus support chatbot

1. State the intended benefit and affected people.
2. Write one realistic failure scenario.
3. Choose one preventive control and one monitoring signal.
4. Define who can review, appeal, or stop the system.

3 minutes individually · 5 minutes in pairs · 2 minutes to share

Module review

1	**Meaning**	Responsible AI connects technical capability with people, context, and consequences.
2	**Lifecycle**	Framing, data, building, evaluation, launch, and monitoring all contain responsible decisions.
3	**Operational pattern**	Value → requirement → control → evidence.
4	**Human role**	Oversight needs authority and information; affected people need recourse.
5	**Practice**	Use purpose, people, failure, control, and evidence questions.



Knowledge check

Three questions to confirm the key distinctions

Question #1: Which action belongs earliest in a responsible AI lifecycle?

A. Add an appeal button after complaints

B. Define the human need and decide whether AI is appropriate

C. Monitor only overall accuracy

D. Assume the training data represents everyone

Answer #1

Correct answer: B. Define the human need and decide whether AI is appropriate

Responsible AI begins by clarifying the need, affected people, success measures, unacceptable outcomes, and whether a lower-risk solution could work.

Question #2: When is a human reviewer a meaningful safeguard?

A. The reviewer sees only the final label

B. The reviewer must approve quickly

C. The reviewer has context, training, time, and authority to overturn the output

D. The system never records disagreements

Answer #2

Correct answer: C. The reviewer has context, training, time, and authority

Human oversight works only when the reviewer can understand the recommendation, question it, change the outcome, and escalate problems.

Question #3: Which sequence best turns fairness into an operational practice?

A. Value → slogan → launch → hope

B. Value → requirement → control → evidence

C. Model → accuracy → deployment → silence

D. Data → model → interface → marketing

Answer #3

Correct answer: B. Value → requirement → control → evidence

A value guides a testable requirement; a control implements it; evidence shows whether the control works in practice.

References: principles and responsible development

Google AI Principles

Current principles: bold innovation, responsible development and deployment, and collaborative progress.
https://ai.google/responsibility/principles/

Responsible AI — Google Cloud

Overview of responsible AI concepts and practical development considerations.
https://cloud.google.com/responsible-ai

AI Safety — Google

Current overview of Google's AI safety research and practices.
https://ai.google/safety/

AI Responsibility Update 2026

Current report on governance, evaluation, safeguards, and deployment practices.
https://ai.google/static/documents/ai-responsibility-update-2026.pdf

References: lifecycle, examples, and activity

AI Responsibility Update 2024

Examples and discussion of responsibility across AI development and deployment.
https://ai.google/static/documents/ai-responsibility-2024-update.pdf

AI Responsibility Update 2026

More recent practices for evaluation, safeguards, and accountability.
https://ai.google/static/documents/ai-responsibility-update-2026.pdf

Introduction to Responsible AI — Google Skills

Approved student activity delivered through the Skills Google learning environment.
https://www.skills.google/course_templates/554

Google AI Principles

Use the live page to confirm the latest wording before teaching.
https://ai.google/responsibility/principles/